

LAB 2

NAME:KATARA NIKUNJ KUMAR KAMLESH KUMAR

ROLL NUMBER=U26AI084

12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

INPUT:

```
#include <stdio.h>

int main(){

char ch;

printf("enter the character value \n");

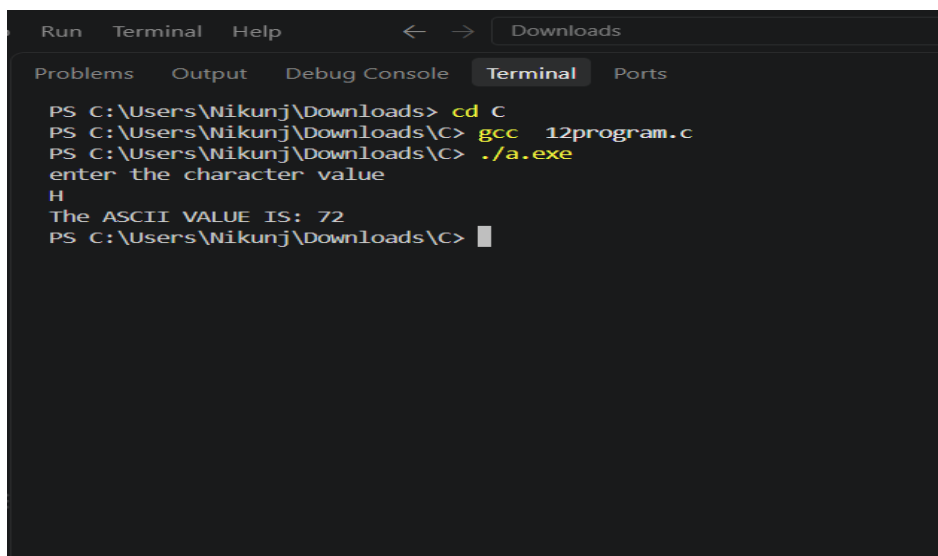
scanf( "%c",&ch);

printf("The ASCII VALUE IS: %d",ch);

return 0;

}
```

OUTPUT



```
Run Terminal Help  < - -> Downloads

Problems Output Debug Console Terminal Ports

PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 12program.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter the character value
H
The ASCII VALUE IS: 72
PS C:\Users\Nikunj\Downloads\C> █
```

13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

INPUT:

```
#include <stdio.h>

int main() {

    int m;

    printf("enter the number: \n" );

    scanf("%d",&m);


    if (m%2==0) {

        printf("the number is even ");

    }

    else {

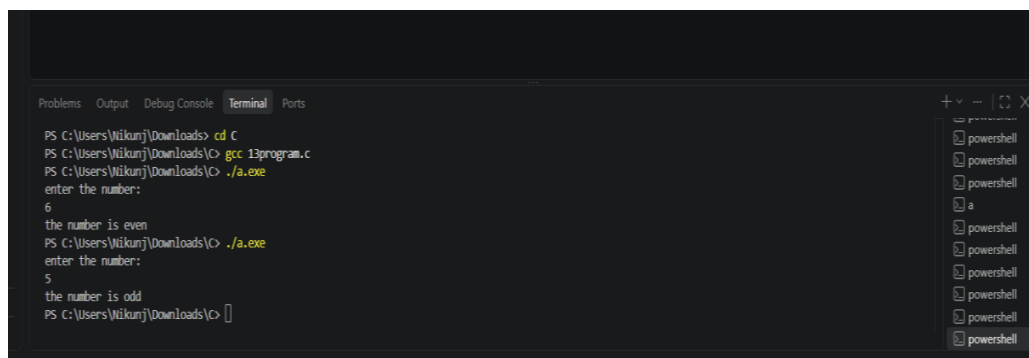
        printf("the number is odd");

    }

    return 0;

}
```

OUTPUT:



The screenshot shows a terminal window with the following content:

```
PS C:\Users\Wilkunj\Downloads> cd C
PS C:\Users\Wilkunj\Downloads> gcc 13program.c
PS C:\Users\Wilkunj\Downloads> ./a.exe
enter the number:
6
the number is even
PS C:\Users\Wilkunj\Downloads> ./a.exe
enter the number:
5
the number is odd
PS C:\Users\Wilkunj\Downloads>
```

The terminal window has tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, showing the command prompt and the execution of the program. The program prompts the user to enter a number, and the user enters 6 and 5. The program outputs "the number is even" for 6 and "the number is odd" for 5.

14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALLCASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

INPUT:

```
#include<stdio.h>

int main() {

    char ch;

    printf("enter the character:");

    scanf("%c",&ch);

    if (ch>='a' && ch<='z' ) {

        printf("it is lowercase \n");

    }

    else if (ch>='A' && ch<='Z') {

        printf("it is upper case \n");}

    else if (ch>='0' && ch<='9') {

        printf("it is a digit \n") ;

    }

    else {

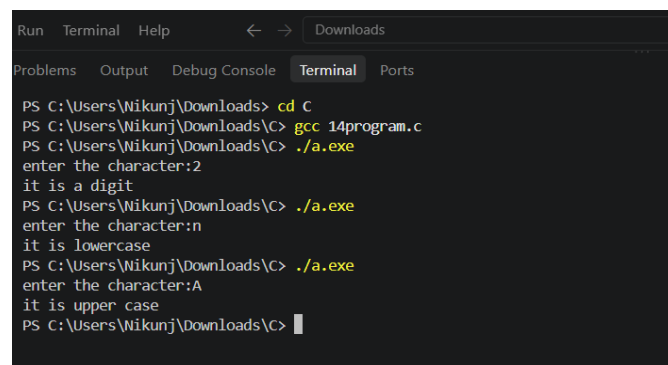
        printf("special symbol \n");

    }

    return 0;

}
```

OUTPUT:



```
Run Terminal Help < -> Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Wikunj\Downloads> cd C
PS C:\Users\Wikunj\Downloads\C> gcc 14program.c
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:2
it is a digit
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:n
it is lowercase
PS C:\Users\Wikunj\Downloads\C> ./a.exe
enter the character:A
it is upper case
PS C:\Users\Wikunj\Downloads\C> 
```

15 PROGRAM TO READ MARKS OF FIVE SUBJECTS AND PRINT DIVISION.

INPUT:

```
#include<stdio.h>

int main() {

    int sub1,sub2,sub3,sub4,sub5,sum,per;

    printf("enter first subject marks: \n");

    scanf("%d",&sub1);

    printf("enter the marks in second subject: \n");

    scanf("%d",&sub2);

    printf("enter third subject marks: \n");

    scanf("%d",&sub3);

    printf("enter fourth subject marks: \n");

    scanf("%d",&sub4);

    printf("enter fifth subject marks: \n");

    scanf("%d",&sub5);

    sum=sub1+sub2+sub3+sub4+sub5;

    per=sum/5;

    if (per>90) {

        printf("A+ \n");

    }

    else if (per>80 &&per<=90) {

        printf("A \n");

    }

    else if (per>70 && per<=80) {

        printf("B \n");

    }

    else if (per>55 && per<=70){

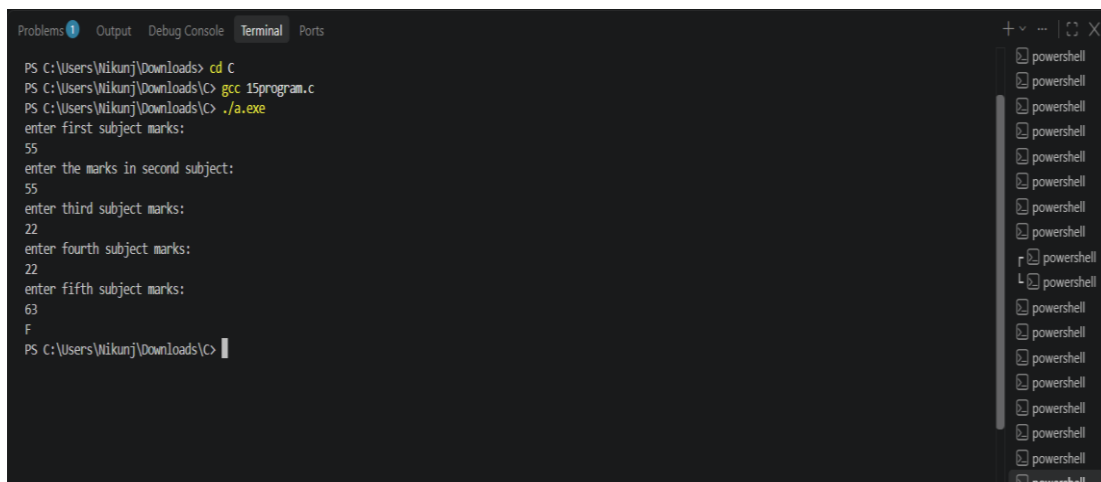
        printf("C \n");

    }

    else if (per>45 && per<=55){
```

```
        printf("D \n");
    }
    else if (per<=45 && per>=0) {
        printf("F \n");
    }
    else {
        printf("invalid input");
    }
    return 0;
}
```

OUTPUT:



```
PS C:\Users\Witkunj\Downloads> cd C
PS C:\Users\Witkunj\Downloads\C> gcc 15program.c
PS C:\Users\Witkunj\Downloads\C> ./a.exe
enter first subject marks:
55
enter the marks in second subject:
55
enter third subject marks:
22
enter fourth subject marks:
22
enter fifth subject marks:
63
F
PS C:\Users\Witkunj\Downloads\C> |
```

16. PROGRAM TO READ THREE NOS. AND PRINT MAX.

INPUT:

```
#include <stdio.h>

int main() {

    printf("please enter distinct numbers! \n");

    int a;

    int b;

    int c;

    printf("enter first number: \n");

    scanf("%d",&a);

    printf("enter second number: \n");

    scanf("%d",&b);

    printf("enter third number: \n");

    scanf("%d",&c);

    if (a>b && a>c) {

        printf(" %d is maximum \n",a);

    }

    else if (b>c && b>a) {

        printf(" %d is maximum \n",b);

    }

    else {

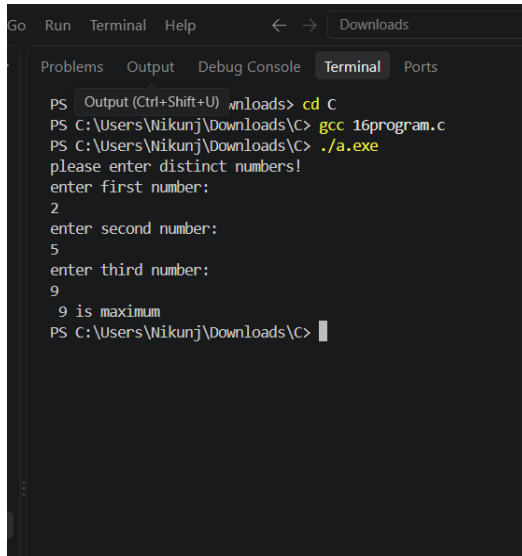
        printf(" %d is maximum",c);

    }

    return 0;

}
```

OUTPUT:



```
Go Run Terminal Help < -> Downloads
Problems Output Debug Console Terminal Ports
PS Output (Ctrl+Shift+U) Downloads> cd C
PS C:\Users\Wikunj\Downloads\C> gcc 16program.c
PS C:\Users\Wikunj\Downloads\C> ./a.exe
please enter distinct numbers!
enter first number:
2
enter second number:
5
enter third number:
9
9 is maximum
PS C:\Users\Wikunj\Downloads\C> |
```

17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

INPUT:

```
#include <stdio.h>

int main() {

    printf("please enter distinct numbers! \n");

    int a;

    int b;

    int c;

    printf("enter first number: \n");

    scanf("%d",&a);

    printf("enter second number: \n");

    scanf("%d",&b);

    printf("enter third number: \n");

    scanf("%d",&c);

    if (a>b && a>c) {

        printf(" %d is maximum \n",a);

    }

    else if (b>c) {

        printf(" %d is maximum \n",b);

    }

    else {

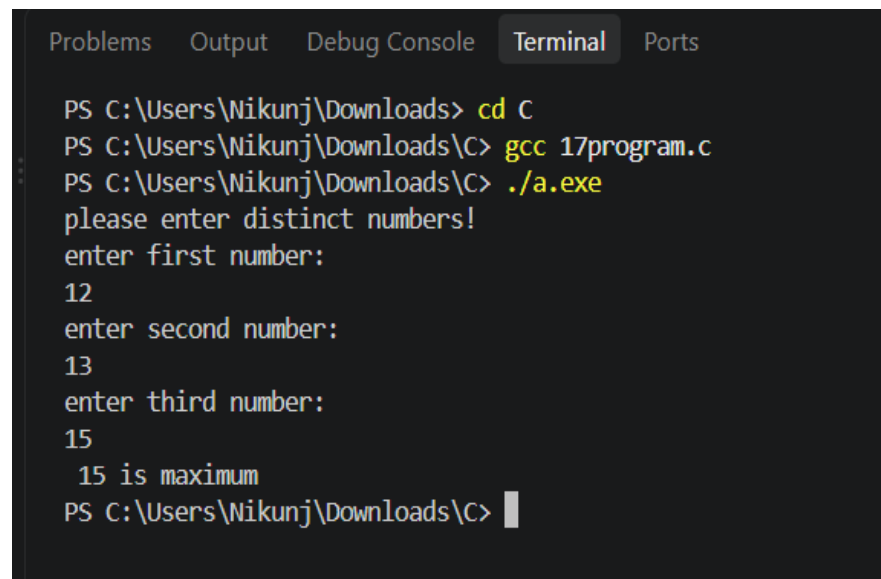
        printf(" %d is maximum",c);

    }

    return 0;

}
```

OUTPUT:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 17program.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
please enter distinct numbers!
enter first number:
12
enter second number:
13
enter third number:
15
15 is maximum
PS C:\Users\Nikunj\Downloads\C> 
```

18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL

(? :)OPERATOR.

INPUT:

```
#include<stdio.h>

int main(){

int a,b,c;

printf("enter a: \n");

scanf("%d",&a);

printf("enter b: \n");

scanf("%d",&b);

printf("enter c: \n");

scanf("%d",&c);

a>b && a>c ? printf("a= %d is maximum",a):printf("-----");

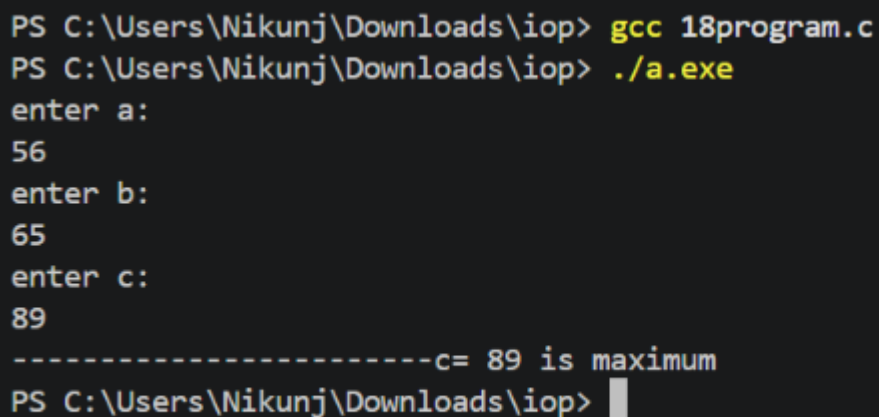
b>c && b>a ? printf("b= %d is maximum",b):printf("-----");

c>a && b>a ? printf("c= %d is maximum",c):printf("-----");

return 0;

}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads\iop> gcc 18program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter a:
56
enter b:
65
enter c:
89
-----c= 89 is maximum
PS C:\Users\Nikunj\Downloads\iop> █
```

19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

INPUT:

```
#include <stdio.h>

int main() {
    char ch;

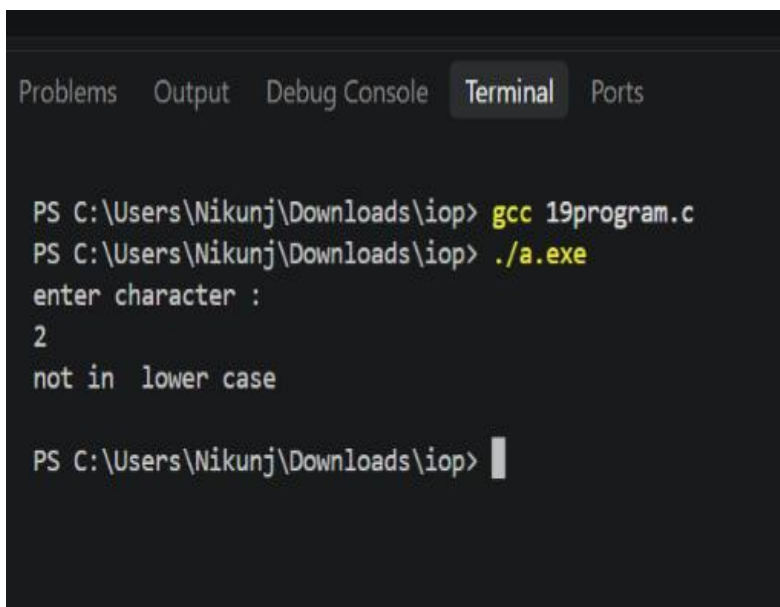
    printf("enter character : \n");

    scanf("%c",&ch);

    ch>='a' && ch<='z'?printf("lower case \n"):printf("not in lower case \n ");

    return 0;
}
```

OUTPUT:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Nikunj\Downloads\iop> gcc 19program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter character :
2
not in lower case

PS C:\Users\Nikunj\Downloads\iop> 
```

20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

INPUT:

```
#include <stdio.h>
```

```
int main() {
```

```
    int a;
```

```
    int b;
```

```
    int c;
```

```
    printf(" enter a ;");
```

```
    scanf("%d",&a);
```

```
    printf("enter b:");
```

```
    scanf("%d",&b);
```

```
    printf("enter code :");
```

```
    scanf("%d",&c);
```

```
    switch(c) {
```

```
        case 1: printf("addition %d",a+b);
```

```
        break;
```

```
        case 2: printf("subtraction %d",a-b);
```

```
        break;
```

```
        case 3: printf("multiplication %d",a*b);
```

```
        break;
```

```
        case 4: printf("division %d",a/b);
```

```
        break;
```

```
        default:printf("invalid code");
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Nikunj\Downloads\iop> gcc 20program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
  enter a ;6
enter b:3
enter code :2
subtraction 3
PS C:\Users\Nikunj\Downloads\iop> 
```

The image shows a terminal window with a dark background. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The terminal content shows a PowerShell prompt 'PS C:\Users\Nikunj\Downloads\iop>' followed by the command 'gcc 20program.c'. The next prompt is followed by './a.exe'. The program then prompts the user for input: 'enter a ;6', 'enter b:3', 'enter code :2', and 'subtraction 3'. The final prompt 'PS C:\Users\Nikunj\Downloads\iop>' is followed by a cursor.

21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR)AND THIS PERFORM USING SWITCH-CASE.(CASE IN CHAR FORM).

INPUT:

```
#include <stdio.h>

int main() {

    char oper;

    printf("tell operation to perform :");

    scanf("%c",&oper);

    int a;

    printf("enter first number :");

    scanf("%d",&a);

    int b;

    printf("enter the second number :");

    scanf("%d",&b);

    switch (oper) {

        case '+' : printf("%d",a+b);

        break;

        case '-' : printf("%d",a-b);

        break;

        case '*' :printf("%d",a*b);

        break;

        case '/' :printf("%d",a/b);

        break;

        default :printf("only + - * / options are available");

    }

}
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 21_program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
tell operation to perform :+
enter first number :5
enter the second number :6
11
PS C:\Users\Nikunj\Downloads\iop> 
```

22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

INPUT:

```
#include <stdio.h>

int main() {

    int n;

    printf("enter N:");

    scanf("%d",&n);

    int factorial=1;

    for (int i=n;i>=1;i--){

        factorial=factorial*i;

    }

    printf("factorial of %d is %d",n,factorial);

    return 0;

}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads\iop> gcc 22program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter N:5
factorial of 5 is 120
PS C:\Users\Nikunj\Downloads\iop> 
```

23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

INPUT:

```
#include <stdio.h>

int main(){
    int base;

    printf("enter base:");

    scanf("%d",&base);

    int power;

    printf("enter power:");

    scanf("%d",&power);

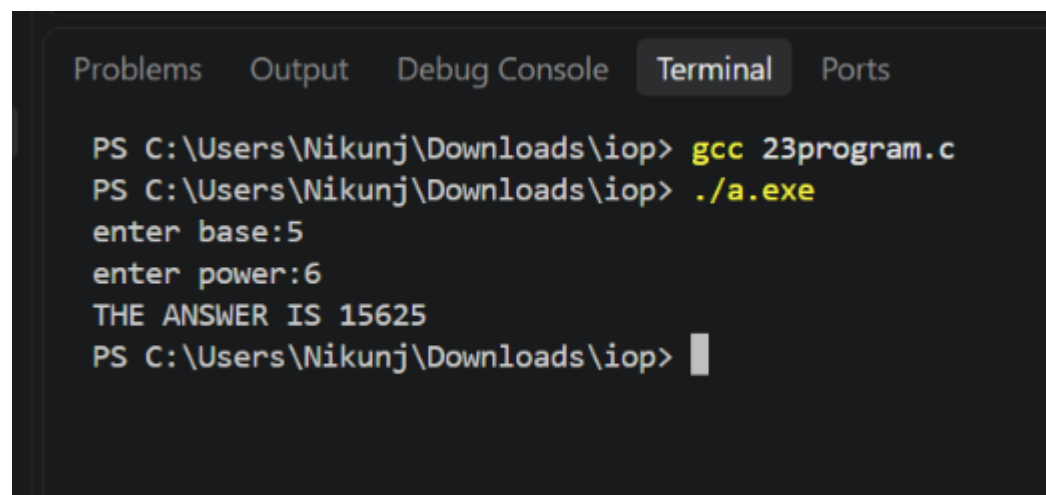
    int ans=1;

    for (int i=1;i<=power;i++){
        ans=ans*base;
    }

    printf("THE ANSWER IS %d",ans);

    return 0;
}
```

OUTPUT:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Nikunj\Downloads\iop> gcc 23program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter base:5
enter power:6
THE ANSWER IS 15625
PS C:\Users\Nikunj\Downloads\iop> 
```

24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

$Y(x,n) = 1+x$ where $n=1$ $1+x/n$ where $n=2$ $1+x^n$ when $n=3$

$1+nx$ when $n>3$ or $n<1$

INPUT:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int n;
```

```
    printf("enter value of n:");
```

```
    scanf("%d",&n);
```

```
    int x;
```

```
    printf("enter value of x:");
```

```
    scanf("%d",&x);
```

```
    int y;
```

```
    if (n==1) {
```

```
        printf("Y= %d \n",1+x);
```

```
    }
```

```
    if (n==2) {
```

```
        printf("Y=%d \n",1+x/n);
```

```
    }
```

```
    if (n==3) {
```

```
        printf("Y=%d \n",1+(int)pow(x,n));
```

```
    }
```

```
    if (n>3 || n<1) {
```

```
        printf("Y=%d \n",1+n*x);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 24program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter value of n:5
enter value of x:6
Y=31
PS C:\Users\Nikunj\Downloads\iop> 
```

25. PROGRAM TO PRINT TABLE OF ANY NO.

INPUT:

```
#include <stdio.h>
```

```
int main(){
```

```
    int i;
```

```
    int n;
```

```
    printf("enter the number :");
```

```
    scanf("%d",&n);
```

```
    for (i=1;i<=10;i++){
```

```
        printf("\n %d * %d = %d \n",n,i,n*i);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
PS C:\Users\Nikunj\Downloads\iop> gcc 25program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter the number :5

5 * 1 = 5

5 * 2 = 10

5 * 3 = 15

5 * 4 = 20

5 * 5 = 25

5 * 6 = 30

5 * 7 = 35

5 * 8 = 40

5 * 9 = 45

5 * 10 = 50
PS C:\Users\Nikunj\Downloads\iop> |
```

26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

INPUT:

```
#include <stdio.h>

int main() {

    int sum1=0,sum2=0;

    int n;

    printf("enter n:");

    scanf("%d",&n);

    for(int i=1,j=2;i<=n,j<=n;i+=2,j+=2){

        sum1=sum1+i;

        sum2=sum2+1;

    }

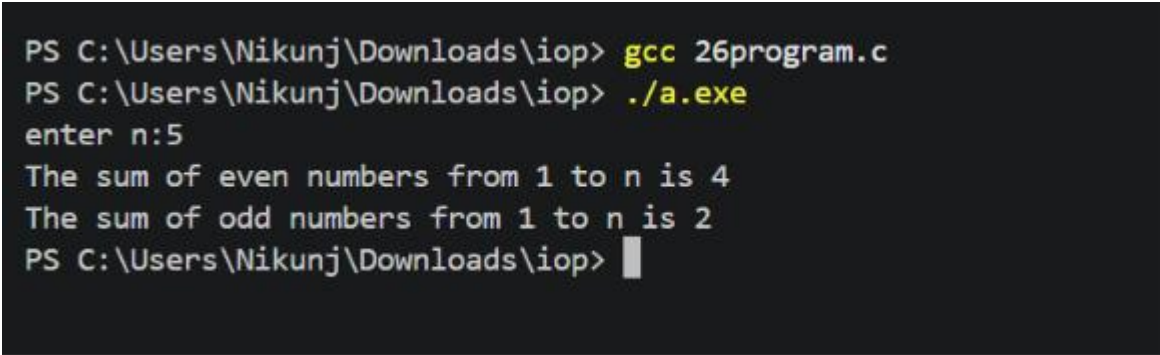
    printf("The sum of even numbers from 1 to n is %d \n",sum1);

    printf("The sum of odd numbers from 1 to n is %d \n",sum2);

    return 0;

}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads\iop> gcc 26program.c
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
enter n:5
The sum of even numbers from 1 to n is 4
The sum of odd numbers from 1 to n is 2
PS C:\Users\Nikunj\Downloads\iop> █
```